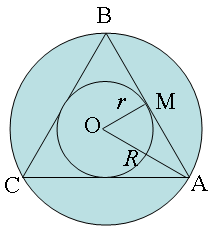
# Team Solutions

1. Let $x = 1.818181\ldots$, then $100x = 181.818181\ldots = 180 + x$, so $99x = 180 \rightarrow x = \frac{180}{99} = \frac{20}{11} \rightarrow N + D = \boxed{31}$.

2. There are 90 two digit numbers. Any power of 2, 3, 5, or 7 will result in one of the four logs to be an integer. There are 7 two-digit powers of these numbers: 16, 32, 64, 27, 81, 25, and 49. The answer is $\boxed{7/90}$.

3. Let $R$ and $r$ be circumradius and inradius of equilateral triangle $ABC$, respectively, and let $M$ be the midpoint of $AB$ (as well as the point of tangency of $AB$ to the incircle). If $O$ is the center of the circles, then $OAM$ is a 30-60-90 triangle. Accordingly, the circumradius is twice that of the inradius. $\pi R^2 - \pi r^2 = 27\pi \rightarrow (2r)^2 - r^2 = 27 \rightarrow 3r^2 = 27 \rightarrow r = 3$. $AM = 3\sqrt{3} \rightarrow AB = 6\sqrt{3}$, so the area of $ABC$ is $(AB^2)\frac{\sqrt{3}}{4} = \left(6\sqrt{3}\right)^2 \frac{\sqrt{3}}{4} = \boxed{27\sqrt{3}}$.




4. Note that in each case the left over pile is four candies short of a full pile. Accordingly, assume Sam was given four additional candies. Then the number of candies he has is a multiple of 6, 9, and 15. The least common multiple of these numbers is 90. Because he has between 100 and 200 candies, Sam must have 180 in his possession. Taking back the 4 additional candies, Sam started with $\boxed{176}$.

5. $JAN$ and $JEN$ are both right triangles, since $JN$ is a diameter of the circle. Accordingly, $JA = 14$ and $JE = 40$. The perimeter of the quadrilateral is $48 + 14 + 30 + 40 = \boxed{132}$.

6. Let $p$ be the probability that HHT appears first. The first time a heads is flipped, if the next flip is also a heads, then HHT must appear before HTT (as there will be a string of two or more heads terminated by a tails flip, forming HHT). If the next flip is tails, then if the following flip is also tails, HTT appears first. If the following flip is heads, then the process repeats itself, and the probability that HHT appears first is $p$. Thus, looking at the two flips after the first heads come up, we get $p = \Pr(\text{H on } 2^{\text{nd}} \text{ flip}) + \Pr(\text{T on } 2^{\text{nd}} \text{ flip, H on } 3^{\text{rd}} \text{ flip})$. Thus $p = 1/2 + (1/4)p \rightarrow (3/4)p = 1/2 \rightarrow p = \boxed{2/3}$.

7. Note that the intersection points $(2, 1)$ and $(8, -1)$ are two of the four corners of a rectangle whose other corners are the points $(a, b)$ and $(c, d)$. In addition, the slopes of the segments joining the corners of the rectangle all have slope $+1$ or $-1$. To find the unknown corners, you can solve the two systems $\{y-1 = x-2, y+1 = -(x-8)\}$ and $\{y-1 = -(x-2), y+1 = x-8\}$ to get the points $(4, 3)$ and $(6, -3)$. Therefore, $a + c = 4 + 6 = \boxed{10}$.

8. By the Angle Bisector Theorem, we have $AD = 3$ and $DC = 4$. Now we find the length of $BD$ by the Law of Cosines. Since angles $ADB$ and $CDB$ are supplementary, their cosines are additive inverses of each other, so we equate $\frac{BD^2+9-36}{6BD} = \frac{64-BD^2-16}{8BD}$ and solve to obtain $BD = 6$. We find the inradius of circle $C_1$ by taking advantage of the formula $Area = rs$, where $r$ is the inradius and $s$ is the semiperimeter of the circumscribed triangle. In the case of triangle $ABD$, we have $\frac{9\sqrt{15}}{4} = r_1 \cdot \frac{15}{2}$, so $r_1 = \frac{3\sqrt{15}}{10}$ and the area of $C_1$ is $\frac{27\pi}{20}$. In the case of triangle $CBD$, we have $3\sqrt{15} = r_2 \cdot 9$, so $r_1 = \frac{\sqrt{15}}{3}$ and the area of $C_2$ is $\frac{15\pi}{9}$. The desired sum is $\boxed{181\pi/60}$.

9. Note that

$$\begin{aligned}
\sum_{n=1}^{1005} \frac{1}{(2n-1)(2n+1)} &= \sum_{n=1}^{1005} \frac{1}{2}\left(\frac{1}{(2n-1)} - \frac{1}{(2n+1)}\right) \\
&= \frac{1}{2}\left(\sum_{n=1}^{1005} \frac{1}{(2n-1)} - \sum_{n=1}^{1005} \frac{1}{(2n+1)}\right) \\
&= \frac{1}{2}\left(1 - \frac{1}{2011}\right) \\
&= \boxed{\frac{1055}{2011}}.
\end{aligned}$$

10. Note that there must be an even number of terms in the sum, and each term must be at least one. Let $T$ be the number of terms in the sum. Note that by subtracting one from each term in an ordered partition of 10, there is a one-to-one correspondence between ordered partitions of 10 consisting of only odd positive integers and ordered partitions of $10-T$ that consists only of non-negative even integers (for example: $3+1+5+1 = 10$ and $2+0+4+0 = 6$). Similarly, there is a 1-to-1 correspondence between ordered partitions of $10 - T$ that consists of only non-negative even integers and ordered partitions of $\frac{10-T}{2}$ that consist of only non-negative integers (for example: $2 + 0 + 4 + 0 = 6$ and $1 + 0 + 2 + 0 = 3$). This set can be counted easily using "stars-and-bars" or your favorite balls-in-buckets counting technique: there are $\binom{\frac{10-T}{2}+(T-1)}{(T-1)}$ ways to do this. Therefore, the answer is

$$\begin{aligned}
\sum_{k=1}^{5} \binom{\frac{10-2k}{2} + (2k-1)}{(2k-1)} &= \sum_{k=1}^{5} \binom{5+k-1}{(2k-1)} \\
&= \binom{5}{1} + \binom{6}{3} + \binom{7}{5} + \binom{8}{7} + \binom{9}{9} = 5 + 20 + 21 + 8 + 1 = \boxed{55}.
\end{aligned}$$

*Note*: An alternate solution is to show that the number of ordered partitions of $n$ using odd positive integers is the $n^{\text{th}}$ Fibonacci number is an exercise left to the reader. (Hint: Split the set of ordered partitions of $n$ into those whose last term is or is not 1.)